

Kartik Marathe

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EDUCATION

Arizona State University, Tempe

Aug 2024 – May 2026

Master of Science — Computer Science

GPA: **4.0 / 4.0**

Selected Coursework: Cloud Computing, Data Processing at Scale, Data Visualization, Mobile Computing

R.V. College of Engineering

Aug 2018 – Jul 2022

Bachelor of Engineering — Electronics & Communication

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, Bash

Backend/Distributed Systems: Spring Boot, Node.js, Flask, REST APIs, GraphQL, gRPC, Kafka, Redis, Apache Solr

Databases/Data Systems: PostgreSQL, MySQL, MongoDB, Neo4j, ETL pipelines, data processing

Cloud/DevOps: AWS (EC2, S3, SQS), Docker, Kubernetes, GitLab CI/CD, Jenkins

Systems/Tools: Linux, Git, GitHub, Jira, Postman, OpenTelemetry, Agile

WORK EXPERIENCE

LERI — Backend Developer — Remote

Apr 2025 – Present

- Scaled a Node.js backend supporting 500+ users, 1,000+ rides, and **10k+ requests/day** with sub-**200 ms** latency.
- Implemented RBAC, rate limiting, Twilio-based 2FA, and OpenTelemetry tracing, reducing fraud attempts by **85%** and cutting debugging time by **60%**.

Arizona State University — Research Aide (Volunteer) — Tempe, AZ

Sep 2025 – Apr 2026

- Created a tech debt analyzer with Python/Flask that processes 200+ files across 5+ repositories, generating automated refactoring recommendations using **15+ complexity metrics** and LLM-assisted code insights.
- Deployed a Dockerized API and tuned PostgreSQL indexing to support sub-second queries across 50k+ code entities, reducing manual review time by **40%** for developer productivity workflows.

The Boeing Company — Associate Software Engineer

Aug 2022 – Jul 2024

- Refactored an EJB desktop application into Spring Boot microservices with CI/CD automation, enabling server decommissioning that saved **\$72k/year** and reduced operational overhead by **80%**.
- Introduced an Apache Solr service indexing 10k+ PDFs with **95%+** relevance accuracy and streamlined GitLab CI/CD pipelines to shrink release cycles from **60** to **10** minutes.
- Maintained 50+ REST endpoints across 4 microservices with **90%** test coverage, supporting 100k+ requests/day, and shipped 5 CMS APIs that improved publishing turnaround by **57%**.

Indian Institute of Science — Research Intern

Jan 2022 – Jul 2022

- Implemented a Python ETL pipeline for 5 sensor streams at **100+ Hz**, enabling BLE-based IoT data capture for clinical research.

PROJECTS

The DOC Parsers

Devpost — GitHub

- Designed a local-first policy intelligence platform with FastAPI, OpenAI, Neo4j, and section-aware PDF retrieval for evidence-backed Q&A, cross-payer comparison, and policy version diffs.

LearnNest

Devpost

- Constructed an AI learning system with FastAPI, Gemini, Pinecone, and Firebase that converts uploaded PDFs into embedding-backed semantic search, personalized quizzes, answer evaluation, and adaptive study notes.

The Mind Shift

Devpost

- Created an AI-powered job matching and resume optimization platform using FastAPI, LangChain, Gemini, Supabase, and TypeScript to score market fit and rank job matches.

EV Energy Flow

GitHub

- Integrated an EV energy orchestration frontend with charger maps, live fleet dashboards, Clerk auth, and FastAPI/WebSocket services for simulation-driven charging workflows.

Secure Task Dashboard

Live Demo — GitHub

- Delivered a MERN task management app with JWT authentication, protected routes, and user-scoped CRUD APIs for secure task creation, updates, and tracking across devices.

Distributed Image Processing

- Assembled a Python/AWS backend with EC2 autoscaling and SQS/S3-based orchestration, supporting 100 concurrent users while processing **10k+ images** with sub-**3 s** latency.

AWARDS & PUBLICATIONS

- Winner — *Innovation Hacks 2025 (ASU)* — Built an AI-powered resume customization platform with ATS scoring.
- Winner — *AEE Hackathon 2026* — EV energy orchestration project for simulation-driven charging and fleet intelligence workflows.
- IEEE INDICON 2022 — Published research on VLSI floorplan optimization using simulated annealing.